
When Are Recursive Language Models Useful? A Cost-Aware, Task-Conditional Evaluation

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Abstract

1 Recursive Language Models (RLMs) handle long inputs through controller-
2 generated programs and additional model calls. Published RLM evaluations
3 typically compare these scaffolds with generic long-context or coding baselines.
4 Deployment requires a different comparison: whether adaptivity improves on
5 routes suited to the task under a common information, scoring and budget contract.
6 We evaluate the released standard RLM against deterministic operators, retrieval,
7 semantic workers and direct inference in an operation-matched audit. On controlled
8 operations and known-family Oolong aggregation, matched routes generally match
9 or outperform standard RLM at lower cost. A source-transfer replication preserves
10 this direction although its clustered exact-rate difference includes zero. Open
11 multi-document search is less stable. An initial untouched-shard replication favors
12 RLM over BM25 but neither a prospectively locked equal-cap comparison nor a
13 post-hoc same-row repeat yields an RLM contrast that survives multiplicity correc-
14 tion. Route ordering changes across these rollouts while RLM remains less reliable
15 and $5.8\text{--}15.9\times$ costlier than the stronger alternatives. Mechanism probes localize
16 errors to operation specificity, output-schema fit, evidence selection and scaffold
17 reliability. These results support reporting recursive inference as a task-conditional
18 accuracy–cost frontier rather than as a single scaffold ranking.

19 1 Introduction

20 Long-context systems can retrieve, compress, apply fixed operators, delegate local judgments,
21 use direct inference or generate adaptive programs. Strong benchmark performance by adaptive
22 programming establishes capability but not deployment value. A deployment decision also depends
23 on the information and tools available to each route, cost, completion and ranking stability.

24 RLMs [Zhang et al., 2025] make this distinction consequential. A controller writes Python over
25 externalized context and may recursively query models. The original paper compares this general
26 route mainly with task-agnostic long-context and coding scaffolds and reports strong OOLONG-
27 Pairs results [Zhang et al., 2025, Bertsch et al., 2026]; follow-ups add self-reflective search, typed
28 combinators and recursive harnesses [Alizadeh et al., 2026, Roy et al., 2026, Lumer et al., 2026].
29 Deployed alternatives include RAG–long-context routing [Li et al., 2024, Chen et al., 2026a], retrieval
30 [Lewis et al., 2020], prompt compression [Jiang et al., 2023], deterministic operators and decomposed
31 workers [Khot et al., 2023]. The relevant comparison is RLM against the strongest affordable route
32 permitted by the task contract.

33 We evaluate this deployment decision directly. For a stated task family, information contract and
34 budget, we ask whether adaptive program generation yields a stable accuracy–cost advantage over
35 operation-matched alternatives. This extends cost-aware agent auditing [Kapoor et al., 2025, Yehudai
36 et al., 2026, Chen et al., 2025b, Bragg et al., 2026] and differs from RLM trajectory diagnosis: the

Table 1: Controlled task-family construction. Each family targets a different long-context operation with a prespecified alternative route.

Family	Operation	N	Expected cheap mechanism
T-STRUCT	Exact pair enumeration	30	SLM extraction + fixed operator
T-MIXED	Directional semantic counting	30	Direct LLM / local judgments
T-NLP (diagnostic)	Standard classification	24	SLM direct

unit is the deployed route, not recursion depth or one scaffold component. Declared specialization, paired uncertainty and rollout stability determine the credibility of the resulting task-conditional route frontier [Yu and Kumbier, 2020].

We compare released standard RLM with deterministic, retrieval, semantic-worker and direct routes. Oolong and an RLM trajectory already decompose labeling from aggregation [Zhang et al., 2025, App. E.2]; our study adds a same-row comparison of complete deployed routes. Table 3 summarizes the boundary. We contribute:

1. An operation-matched protocol that fixes rows and scorers, declares route-specific prior knowledge and reports accuracy, failures, latency, calls and cost. An answer-contract repair separates finalization failure from route quality.
2. Wrong-family, same-operator, finalization and trace controls that distinguish operation knowledge from tool access and localize differences in semantic judgment, output schema and scaffold reliability; prior RLM trajectory diagnoses do not combine these controls [Zhang et al., 2025, Wang, 2026].
3. A task-conditional deployment boundary: known-operation routes lead on Oolong under paired tests, while BrowseComp route ordering changes across same-row rollouts and no RLM contrast survives Holm correction. The resulting two-tier recommendation shows why a single scaffold score is insufficient for deployment.

2 Operation-Matched Evaluation

Scope. The audited system is the official `rlms v0.1.3` standard inference-time RLM: a controller generates Python over externalized context and may call additional models before answering. We audit this public, installable implementation against operation-matched alternatives on the same instances. Direct standard-RLM accuracy rows cover T-STRUCT, T-MIXED and the Oolong slices; T-NLP is a compatibility diagnostic. Claims remain scoped to this implementation; tuned depth, reasoning or stronger-family variants may move the frontier. We also run MIT OASYS’s official post-trained RLM-Qwen3-30B-A3B policy and the unmodified public λ -RLM code with its Qwen3-8B tier on frozen CD-15 under the same input/scorer contract. These are same-contract diagnostics, not reproductions of the official SRLM, λ -RLM or RAH benchmark protocols; adapter-based rows remain pressure tests only.

Tasks. Table 1 defines three controlled operation families. Public evidence adds LongBench-v2 (LB2) Academic QA ($N=94$) [Bai et al., 2025], label-provided Oolong-synth ablations ($N=80, 150$), standard-unlabeled Oolong slices ($N=30, 120, 150, 45$) and BrowseComp-Plus estimates on an original frozen $N=80$ manifest, a stopped attempted- $N=49$ replication and two fixed- $N=45$ untouched-query endpoints [Chen et al., 2025a]. The second 45-row endpoint prospectively adds iterative lexical search and equal decision/time caps across all four routes. SU-150 spans 8k–131k tokens but overlaps previously evaluated windows; CD-45 is balanced over 1k, 4k and 262k tokens and excludes prior IDs/windows, while VC-45 uses the official validation split with two upstream corpora absent from all previous Oolong runs. Oolong exposes no source-document ID so neither 45-row slice supports document-disjointness claims.

Table 2: Model roles and their corresponding evaluation settings.

Role	Model	Use
RLM ctrl.	Gemini 2.5 Flash; GPT-5; Qwen3.6-35B-A3B	Standard scaffold; direct matched Gemini; BrowseComp matched routes
Trained/code RLM	Qwen3-30B-A3B; λ /Qwen3-8B	Frozen CD-15 diagnostics
Direct LLM	Gemini 2.5 Flash; Qwen3.6-35B-A3B	Direct-context and matched-controller baselines
SLM worker	Gemma 4 26B-A4B	Direct, CoT, labeling and fallback routes
Cheap controller	GPT-4o-mini	Deliberately weak cheap controller; T-NLP orchestration only

Evaluation design. The controlled families cover exact aggregation, semantic counting and classification; label-provided Oolong separately probes output-schema inference. We then evaluate public slices chosen to probe settings in which RLM should be competitive. This design trades leaderboard breadth for trace-level analysis: headline rows are tied to frozen inputs, cost logs, contract labels and leakage/split checks. Figure 1 declares what each route is allowed to know before any comparison is run. The non-RLM routes intentionally use task-family prior knowledge, fixed operators and prompt/schema engineering; we treat this as deployment specialization to account for, not as a scaffold-only causal intervention. Our recommendation would be overturned by an official trained policy or same-contract SRLM/ λ -RLM/RAH reversal, a wrong-family typed route that still wins or an RLM route with the same operator library matching cost and accuracy.

Baselines and model roles. Table 2 lists the role each model plays. Baselines include BM25, iterative lexical search, textual decomposition, fixed operators, Gemma 4 26B-A4B workers, LLM+SLM routes, direct Gemini 2.5 Flash and standard RLM. The primary Oolong RLM controller is Gemini 2.5 Flash; T-STRUCT uses GPT-5 and the BrowseComp-Plus matched-controller audit uses Qwen3.6-35B-A3B for all routes. Appendix A reports current-controller sensitivities and official-policy/code diagnostics. Within each endpoint all routes share one generation contract. The fixed- $N=45$ and prospective equal-cap BrowseComp routes use Qwen model-card settings with seed 20 and thinking disabled to keep decoding policy matched under the common caps; the post-hoc repeat changes only the seed to 21. The $N=80$ manifest and both judges use temperature 0, while GPT-5.6 rows omit unsupported temperature. Self-hosted equal-cap routes are pinned by model revision and container digest and hosted models by dated slug. Exact settings and identifiers are reported in Appendix H. Costs use uncached list prices and token logs; they exclude engineering effort and successful-completion totals can understate attempted scaffold billing.

Protocol and evidence tiers. Each headline claim specifies task, baseline and scorer. Controlled families establish mechanisms under identical generators; public Oolong and BrowseComp rows test transfer; LongBench-v2 diagnostics probe context limits, MCQ position, schema rigidity and split overlap. We label each public row as validation, ablation, replication or pilot under its stated contract. Appendix H records the model, protocol, compute and release boundaries needed to interpret the evidence.

Estimands, scoring and uncertainty. For paired routes r, s on n rows, $\hat{\Delta}_A = n^{-1} \sum_i (Y_{ir} - Y_{is})$ and $\hat{\Delta}_C = n^{-1} \sum_i (C_{ir} - C_{is})$; failed or invalid executions have $Y = 0$. Non-dominated routes form the observed task-conditional frontier. RLM is dominated when $\hat{\Delta}_A \leq 0$ and $\hat{\Delta}_C > 0$; positive tradeoffs require deployment-specific valuation. Paired accuracy uses McNemar tests (exact for small discordance); b, c denote first-only and second-only correct. Difference intervals use fixed-seed percentile bootstraps with 50,000 paired-row draws for BrowseComp and 5,000 context-window-cluster draws for Oolong. Primary controlled estimands are paired accuracy and captured cost

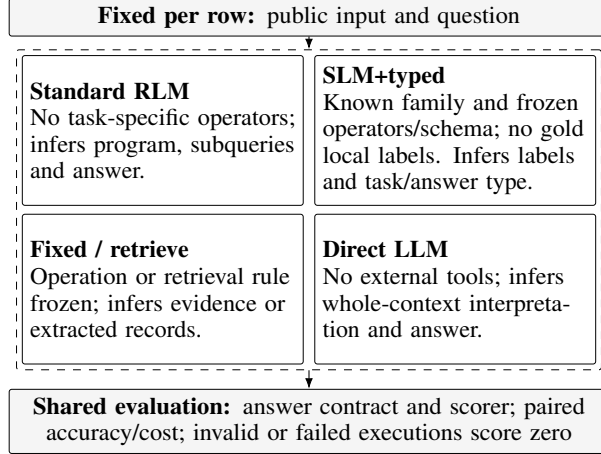


Figure 1: Route and information contracts. Allowed deployment prior knowledge is declared before comparison. The estimates compare complete routes rather than isolating a scaffold-only causal effect, and they do not evaluate an automatic router.

on T-MIXED and standard-unlabeled Oolong. BrowseComp-Plus supplies separate open-search estimates: the original $N=80$ manifest is bounded by two-episode completion and post-stop visibility; an attempted- $N=49$ endpoint stopped score-blind; an initial replication fixes $N=45$ before output; and the strongest prospective endpoint freezes 45 further untouched queries with iterative search and matched decision/time caps, following matched-budget practice [Alizadeh et al., 2026]. Failures are incorrect in every fixed denominator. Subgroup, controller and mechanism-probe tests are descriptive; we make no global familywise claim. The equal-cap endpoint applies Holm adjustment over RLM versus BM25, textual decomposition and iterative search. Oolong scores exact matches and accepted relative-frequency substrings, with $0.75^{|\Delta|}$ for numeric near-misses; we report both the binary exact rate and the partial-credit mean score. MCQ predictions are mapped back after answer shuffling. Because Oolong questions can share contexts, uncertainty resamples 72, 73, 32 and 20 clusters for $N=120$, SU-150, CD-45 and VC-45. Intervals are conditional on observed executions; a full CD-45 repeat documents rollout sensitivity for that frozen Oolong endpoint.

Cost accounting. RLM cost is decomposed into controller generation, child-query execution and final extraction. API totals capture successful completions and are lower bounds where failed calls return no usage. Direct and SLM baselines are single-call unless retrieval-augmented; fixed operators have zero marginal model cost but nonzero engineering cost. At SU-150, unrounded observed API savings are $\$0.022040733/\text{case}$ ($\$0.022041$ rounded), so a hypothetical $\$1,000$ implementation budget breaks even after $\lceil \$1,000/\$0.022040733 \rceil = 45,371$ cases [Kapoor et al., 2025]. This is an amortization sensitivity, not a labor estimate or TCO. Self-hosted BrowseComp costs use active runtime by route with generation-plus-judge total reported separately. Reported statistics and cost totals are computed from stored logs.

3 Results

Matched routes lead on known operations. On T-STRUCT, Gemma extracts records and a fixed Python operator performs the aggregation, which is exact by construction, so this row measures extraction success and cost. It reaches 100% accuracy versus 96.7% for RLM at about $99\times$ lower captured deployed-configuration API cost; because both are near ceiling, the result establishes cost dominance only, and supports neither a reliable accuracy difference nor a pure matched-model scaffold-overhead estimate. T-MIXED requires semantic judgments. A matched Gemini ablation separates finalization from semantic error: the count-only finalizer returns no valid answers while the JSON finalizer returns valid answers on all 20 examples and solves 13. A frozen $N=30$ endpoint compares direct Gemini with JSON-finalized RLM: direct Gemini solves 28/30 at $\$0.0223$, RLM solves 18/30 at $\$0.1089$ (paired exact $b=12$, $c=2$, $p=0.0129$) and remains about $4.9\times$ costlier. We treat T-NLP only as an interface diagnostic because no official RLM accuracy run is logged; moreover,

Table 3: Main results and mechanism boundaries by evidence tier. LP entries are mean/exact; others are exact. “Best non-RLM” prioritizes accuracy on T-MIXED. T-STRUCT compares deployed configurations; CD-45 is diagnostic. Equal-cap BrowseComp route-attributed active-runtime cost is \$13.279 (\$15.015 with interrupted-episode lower-bound billing and judging).

Regime	Best non-RLM	Standard RLM	Boundary supported by this row
T-STRUCT exact aggregation	Gemma extract + fixed agg.: 100%, \$0.0002/ex	96.7%, \$0.0198/ex	Known exact aggregation: a typed pipeline matches/exceeds RLM at much lower captured deployed-configuration API cost (about 99 $\times$ in this cross-model run; still 9.9 $\times$ if Gemma extraction is repriced 10 $\times$ higher).
T-MIXED directional count	Direct Gemini: 28/30, \$0.0223 total	RLM: 18/30, \$0.1089 total	Same-model direct inference wins on the frozen $N=30$ endpoint; the $N=20$ finalizer ablation isolates answer-contract failure.
Oolong specialization stress (CD-45; 1k/4k/262k)	Correct-family SLM+typed: 36/45; wrong-family: 2/45	Same-tools RLM: 31/45 or 29/45; GPT-5/GPT-5.6 Luna: 23/45, 26/45; at least 2.3 $\times$ costlier	Correct-family knowledge is necessary; same tools or tested controller changes narrow but do not reverse the ordering.
Oolong label-provided LP-80/LP-150	Typed ops: 100% / 97.3%	RLM: 82.4% / 84.7%	RLM is strong under flexible output schemas but gold-label typed operators expose the schema ceiling.
Oolong standard-unlabeled (SU-150; VC-45)	SLM-labeler + typed ops: 98/150, \$0.839; unused-source VC-45 35/45, \$0.312	RLM 68/150, \$4.145; VC-45 28/45, \$1.139	Five 8k–131k buckets plus pre-output-locked source-corpus transfer: SLM+typed has the best accuracy/cost point; VC-45’s exact-rate interval includes zero while its mean-score interval does not.
BrowseComp prospective equal-cap ($N=45$)	Text: 37/45, \$1.400; iterative: 33/45, \$1.710; BM25: 25/45, \$0.296	RLM: 31/45, \$9.873	The earlier RLM–BM25 gain does not transport: no RLM contrast survives Holm correction, and RLM is 5.8–7.1 $\times$ costlier than the stronger adaptive routes.

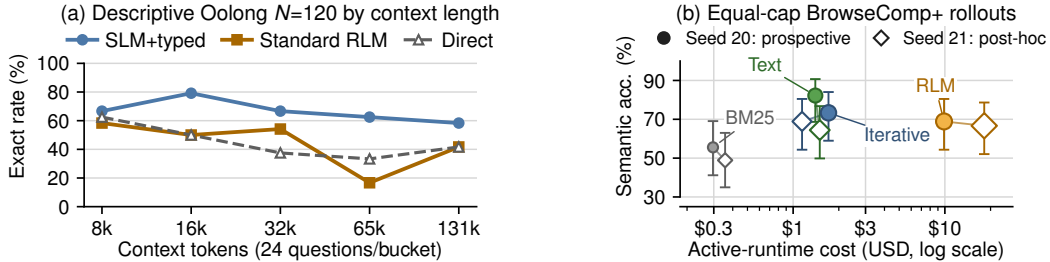


Figure 2: Accuracy and cost across task regimes. (a) Descriptive exact rates on standard-input Oolong development data ($N=120$; 24 per bucket). The inferential contrast is the paired cluster-resampled difference reported in the text, not a per-cell interval. (b) Accuracy versus active-runtime cost on the matched-Qwen equal-cap endpoint (filled circles: prospective seed 20; hollow diamonds: same-row post-hoc seed 21; $N=45$ each). Whiskers are Wilson intervals; marker area encodes all-row calls (RLM/iterative/text/BM25: 476/426/311/42 and 667/276/312/42). No RLM contrast survives Holm correction in either rollout.

146 the classification target requires categorical labels while the current aggregation scaffold assumes
147 numeric-style answers.

148 **Open-search advantages do not survive equal caps.** BrowseComp-Plus provides natural open
149 search without a known typed operator and already reports retriever/agent separation and cost
150 analyses; we add operation-matched routes under equal caps on frozen shards. The original frozen
151 $N=80$ manifest uses 1,000 documents per query and matched-Qwen routes: standard RLM, BM25
152 and textual decomposition. It was completed across two disclosed episodes after ranks 1–37 became
153 score-visible. A method-blinded Gemma 4 judge, sharing no model family with any judged route,
154 scores semantic equivalence to the benchmark’s gold answer and gives RLM 54/80 (67.5%), text

155 48/80 (60.0%) and BM25 39/80 (48.8%); only the 18.8-point RLM–BM25 interval excludes zero.
156 A separate GPT-5.6 blind audit likewise supports RLM over BM25 while leaving the RLM–text
157 comparison uncertain.

158 In a fixed- N replication, we froze 45 rows from a different official query shard with zero overlap
159 against earlier manifests. Model, routes, scorer and failure accounting were unchanged. Gemma 4
160 scores RLM 33/45 (73.3%), text 29/45 (64.4%) and BM25 21/45 (46.7%). RLM exceeds BM25 by
161 26.7 points (paired 95% interval 11.1–44.4; Holm $p=0.0151$), while its 8.9-point lead over text is
162 uncertain (−6.7–24.4). A blinded GPT-5.6 Sol audit agrees on 128/129 successful outputs ($\kappa=0.983$).
163 RLM completes 42/45, costs \$6.746 and has a 425.7-second p95 latency, versus 45/45, \$1.610 and
164 44.9 seconds for text.

165 The prospective equal-cap endpoint changes this ordering. We froze 45 untouched queries, added
166 iterative lexical search and attempted all 180 Qwen3.6 route cells under common 20-decision/1,200-
167 second caps before semantic scoring. A rank-19 interruption was recovered under the frozen
168 hash-bound pre-score protocol. RLM scores 31/45 (68.9%), versus iterative 33/45 (73.3%), text 37/45
169 (82.2%) and BM25 25/45 (55.6%). Differences are −4.4, −13.3 and +13.3 points, with discordances
170 6/8, 5/11 and 10/4. Paired 95% intervals are [−20.0, 11.1], [−31.1, 4.4] and [−2.2, 28.9] points; no
171 contrast survives Holm correction ($p_{\text{Holm}} \geq 0.539$). Sol/primary agreement by route is 41/41, 45/45,
172 45/45 and 41/42 (RLM/iterative/text/BM25; $\kappa=0.985$).

173 RLM completes 41/45 with 50.1/628.8-second median/p95 latency. Iterative/text complete all 45
174 with 14.1/90.2 and 22.1/47.6 seconds; BM25 completes 42/45. On 39 common completions, scores
175 are 30/28/31/22; this post-treatment sensitivity leaves the adaptive routes near parity. A question-only
176 Qwen leakage probe scores 0/45 under both judges: neither finds a correct answer without the corpus.
177 The earlier RLM–BM25 advantage therefore does not transport; Figure 2(b) plots both equal-cap
178 rollouts against active-runtime cost. Open search alone is not a sufficient routing signal.

179 A post-hoc seed-21 repeat reaches the same inferential conclusion. RLM/iterative/text/BM25 com-
180 plete 37/45, 45/45, 45/45 and 42/45 and score 30/31/29/22; no RLM contrast survives Holm
181 ($p_{\text{Holm}} \geq 0.346$). A cap-valid completion finishes within 1,200 seconds: RLM has 36/45, and
182 on 34 rows cap-valid for every route the scores are 28/23/20/18. This post-treatment sensitivity puts
183 RLM first, separating attrition from conditional accuracy. Across four descriptive cross-seed tests,
184 RLM is stable ($b=8, c=7, p=1.0$) while text shifts ($b=9, c=1, p=0.021$); the latter does not survive
185 four-test correction. RLM costs 12.1–15.9× the adaptive routes. Sol agrees on 166/169 outputs
186 ($\kappa=0.961$); the repeat is not pooled with the prospective endpoint.

187 **Oolong specialization persists across labels and sources.** On label-provided LP-80, RLM reaches
188 82.4% mean score against 70.0% for a generic fixed-parser/SLM fallback, although that difference
189 is uncertain ($p=0.089$); its largest gain is on timeline questions and deterministic parsing remains
190 perfect on counting. Gold-label typed operators reach 100% on the same split, so RLM is not
191 the strongest LP-80 route. On frozen held-out LP-150, RLM reaches 84.7% exact while typed
192 rules fixed before evaluation reach 97.3% ($b=23, c=4, p<0.001$). Gold local labels make these
193 aggregation/output-schema ablations, not end-to-end Oolong results. They show that RLM handles
194 flexible schemas better than a generic parser but not that open-ended Python is uniquely required
195 when the schema family is known.

196 All central standard-unlabeled checks use context_window_text without gold local labels. An
197 $N=120$ development replication established the SLM>RLM direction across every 8k–131k bucket
198 (Figure 2(a)), with a paired 22.5-point exact-rate advantage ($b=37, c=10, p=9.8 \times 10^{-5}$; 72-cluster
199 interval 11.9–33.0). The fixed pipeline was evaluated on SU-150, CD-45 and pre-output-locked
200 VC-45; before scoring, one parser repair was applied only to rows returning no call, a condition
201 visible without labels.

202 On SU-150, SLM+typed solves 98/150 at \$0.839, versus RLM 68/150 at \$4.145 and direct Gemini
203 65/150 at \$1.939. Discordance favors SLM over RLM ($b=44, c=14, p=1.00 \times 10^{-4}$); no direct–RLM
204 difference is detected ($p=0.788$). The 73-cluster interval gives a 10.6–29.9 point SLM advantage but
205 all rows reuse previously evaluated context windows so SU-150 is scale/method-lock evidence only.

206 VC-45 removes the known source-corpus overlap: its official validation rows come from spam
207 and trec_coarse, absent from the test split and all previous Oolong evaluations. The question
208 parser agrees with metadata on all 45 VC-45 rows; SU-150 and CD-45 carry no equivalent check.

SLM+typed solves 35/45 at \$0.312, versus RLM 28/45 at \$1.139 and direct Gemini 23/45 at \$0.658. SLM–RLM discordance is $b=10, c=3, p=0.092$ (Holm $p=0.185$); the 20-cluster exact-difference interval is 0.0–31.1 points while the mean-score interval is 3.0–30.9. RLM reaches 13/15 on user questions but 6/15 on counting. Task and source are not factorially balanced, so subgroup differences are descriptive. The specialized route remains cheaper and directionally favored under source-corpus transfer but VC-45 alone establishes neither a significant exact-rate advantage nor document-level independence.

CD-45 retains the specialization controls: wrong-family typing collapses from 36/45 to 2/45 while giving free-form standard RLM the same operator library reaches 31/45 or 29/45 but remains at least $2.3\times$ costlier. An independent standard-RLM repeat drops from 27/45 to 21/45. Together, SU-150 supplies scale, VC-45 supplies source-corpus transfer and CD-45 isolates specialization and same-tools behavior; none is a full Oolong leaderboard.

The Oolong ordering persists across tested controllers. The primary Oolong evaluation uses Gemini 2.5 Flash with the released standard-RLM stack and complete cost logs. Strict-stop GPT-5 and GPT-5.6 Luna standard-RLM runs on CD-45 reach 23/45 and 26/45; SLM+typed remains higher ($p=0.00235$ and $p=0.041$). On frozen CD-15, the official trained Qwen policy reaches 10/15 and the prospectively frozen official-code λ -RLM row reaches 8/15, versus 15/15 for SLM+typed. The complete λ row attempts all IDs on one pinned Qwen3-8B/vLLM deployment and counts two 1,800-second timeouts incorrect. Paired exact outcomes favor SLM+typed ($b=7, c=0, p=0.0156$); no difference is detected against direct Gemini, standard RLM/Gemini or official Qwen (all $p\geq 0.625$). Across controllers and policies, no sensitivity reverses the SLM+typed ordering; different manifests, denominators and fidelity tiers preclude pooling. These checks support controller robustness of the Oolong ordering within the evaluated contracts. The CD-15 diagnostics are not reproductions of the official SRLM/ λ -RLM/RAH benchmark protocols, so the central claim remains scoped to the released standard implementation.

The observed frontiers imply a task-conditional routing rule. Evidence supports two tiers. When task type is known or reliably inferred from the query, test the cheapest mechanism that matches the operation: fixed operators for exact aggregation, retrieval or semantic workers for matching, SLM/direct calls for local classification and question-parsed typed aggregation for known Oolong-style aggregation families. When no tested typed route captures the request, compare retrieval, textual decomposition and iterative search before treating RLM as a candidate. The Oolong results show that flexible schemas need not require open-ended generated Python; the prospective BrowseComp endpoint shows that open search does not itself justify RLM.

4 Mechanism Analysis

Observed failures span task interpretation, evidence selection, aggregation, tool use and finalization rather than a single RLM component. Appendix C.1 organizes these observations into known-operator overhead, final-answer contract sensitivity, semantic-local errors, output-schema rigidity, open-search instability, context-window constraints and multiple-choice format bias.

LLM+SLM helps with semantic decomposition, not fixed operations. The evaluation also tests the intuitive alternative "use a strong LLM to orchestrate cheap SLM workers" [Narayan et al., 2025]. This helps when the controller mainly decomposes semantic work but it is not an automatic replacement for deterministic operators. On exact tasks, a controller is unnecessary because the program is known. On T-NLP, LLM+SLM reaches 83.3%, compared with 87.5% for SLM direct while costing about $3.4\times$ more within the evaluated route set. On LB2-style QA, it inherits retrieval and MCQ-bias errors unless the evidence-selection stage is corrected. The viable version is therefore task-conditioned: use LLM+SLM for local interpretation, not for operations already solved by fixed code or retrieval.

Traces expose failures at distinct stages. Four reported CD-45 cases in Appendix C.2 cover a relation error, a direct whole-context error, a typed tie/normalization edge and same-tool interface overhead. Stored T-MIXED traces likewise show successful dual-gate filters but also vocabulary overreach and brittle finalization. These traces localize where failures surface; they do not attribute them to the scaffold as a whole.

Multiple-choice position bias limits the QA diagnostic. LongBench-v2 Academic QA is multiple-choice so answer-position priors confound retrieval-vs.-RLM comparisons. On the $N=50$ discovery set, canonical BM25 predicts A 68% of the time while gold-A is 18%. Rotating priming improves pooled $N=94$ ($p=0.015$) but not held-out $N=44$ ($p=1.0$); single-call accuracy is unmeasured. We therefore keep it as a validity diagnostic, with prior debiasing comparisons in Appendix B.

5 Related Work

RLM externalizes context through controller-generated programs and recursive calls [Zhang et al., 2025]. Follow-ups question recursion as the driver or add typed operators, coding agents and recursive harnesses [Alizadeh et al., 2026, Roy et al., 2026, Ehrlich and Blackman, 2026, Cao et al., 2026, Lumer et al., 2026]. Wang finds task-dependent depth, cost and strong-model degradation [Wang, 2026]. Our operation-matched comparison fixes rows and scorers, reports paired uncertainty and treats the deployed route rather than recursion depth as the unit of analysis.

Cost-aware scaffold evaluation shows that simpler baselines can Pareto-dominate agents [Kapoor et al., 2025]. This study applies that principle to complete long-context routes under explicit information, scoring and budget contracts and measures recommendation stability across rollouts.

Oolong separates local labeling from aggregation [Bertsch et al., 2026]; its labels support our mechanism ablations. Long-context suites test realistic reasoning, effective context and retrieval/length effects [Bai et al., 2025, Chen et al., 2026b, Hsieh et al., 2024, Yen et al., 2025, Zhang et al., 2024, Modarressi et al., 2025, Li et al., 2025]. Decomposition, search and routing systems select evidence or execution routes [Xu et al., 2026, Song et al., 2026, Ong et al., 2025, Li et al., 2024, Chen et al., 2026a]. Corpus-scale evidence shows that lexical and agentic search can reverse order as the search space grows [Wang et al., 2026]. We compare fixed, worker and RLM routes under logged cost and explicit evidence grades.

6 Limitations

External validity. The study diagnoses task-conditional failure modes, not average long-context performance. VC-45 is source-corpus but not document disjoint because Oolong lacks IDs; SU-150 reuses windows. Official-family evidence covers CD-15. The controlled curve ends at 131k; longer rows are stress tests.

Measurement. Equal-cap evaluation attempts all 180 cells under fixed caps across three execution episodes. Evaluation validity is assessed through cross-family judge agreement and a question-only leakage control. The study does not include human adjudication, so these checks establish stability rather than human validity amid contextual judge fragility [Tan et al., 2025, Xu et al., 2025].

Cost and scope. Costs exclude engineering and TCO; alternatives are not exhaustive and recursion depth is fixed. BrowseComp bins reach $n=10$, so we infer no subgroup effects. A same-row embedding rerank remains below RLM but is not full-corpus dense retrieval (Appendix B). Seven repeat RLM calls exceed the cap and score incorrect. Self-hosted and API dollars remain separate. Automated route choice could suppress useful methods in untested domains.

Ethics and AI assistance. Public or synthetic benchmarks involve no human subjects. AI tools assisted with coding, experiment orchestration and language editing; the authors verified the analyses and manuscript.

7 Conclusion

Matched routes lead on known operations. Across two equal-cap rollouts no RLM contrast survives multiplicity correction while RLM costs $5.8\text{--}15.9\times$ more. RLM evaluation should therefore report a task-conditional route frontier, not promote a single scaffold ranking into a deployment policy.

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431 A Protocol Details

432 **Cost-aware audit protocol.** Table 4 summarizes the protocol used throughout this study. Because 433 the audited object is a scaffolded inference procedure, a row should not be promoted from diagnostic 434 evidence to method evidence unless the input contract, output schema, scorer, failure accounting and 435 cost trace are matched.

436 **Role-based model coverage.** The primary panel assigns models to deployment roles rather than 437 constructing a broad leaderboard: Gemini 2.5 Flash controls the logged standard RLM and supplies 438 direct frontier inference, Gemma 4 26B-A4B supplies SLM worker paths and GPT-4o-mini controls 439 the short T-NLP LLM+SLM route. GPT-5 is used for the T-STRUCT frontier-controller exception 440 and a CD-45 controller sensitivity. Many central alternatives are deterministic or structural so the 441 unresolved breadth question is same-contract coverage of stronger RLM-family systems rather than 442 additional direct-LLM columns.

443 That stronger-family evidence includes frozen-manifest adapter pressure tests, matched standard- 444 unlabeled smokes, the CD-15 repeatability study and CD-45 same-operator-library diagnostics. 445 The unmodified public λ -RLM repository attempts all 15 frozen CD-15 IDs with its published 446 Qwen3-8B settings and reaches 8/15 exact with two declared timeouts. MIT OASYS’s official 447 `r1m-qwen3-30b-a3b-v0.1` post-trained policy reaches 10/15 exact with one timeout through the 448 canonical `r1m` harness and model-card Oolong settings. The λ row is official-code cross-task evidence 449 rather than an original-paper reproduction; rankable official SRLM and same-contract RAH rows 450 remain unavailable.

451 Controller-freshness and compatibility checks are reported separately from family parity. Table 5 452 gives the contracts for the complete central and sensitivity rows; these checks do not constitute full 453 official SRLM/ λ -RLM/RAH benchmark parity.